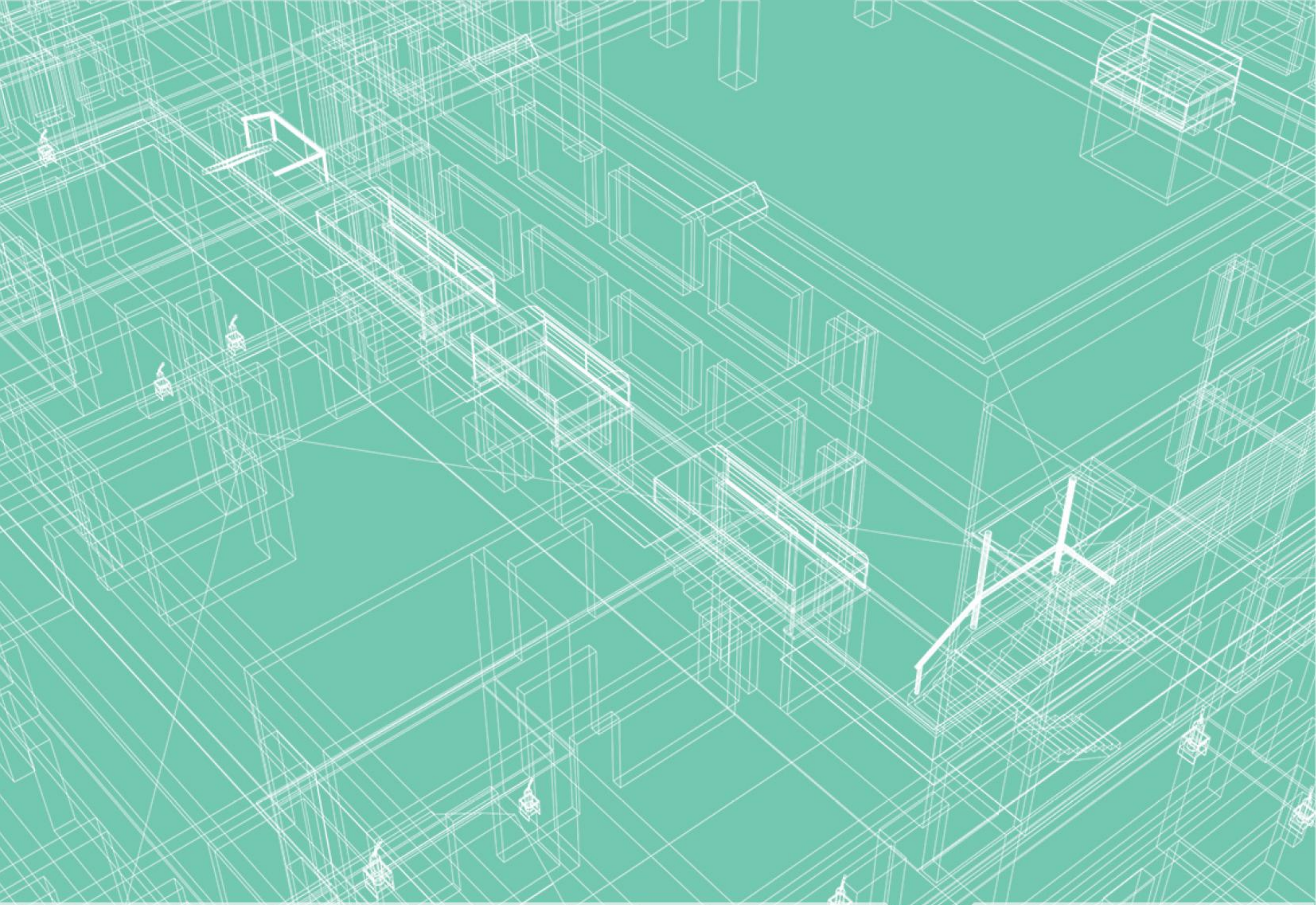

# SWITCHYARD ROAD STANWELL QLD

# PERFORMANCE SOLUTION – ROOFWATER DRAINAGE

# BOX GUTTER DESIGN.

NO: Q250128

JHA

JHASERVICES.COM

**DISCLAIMER**

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# Document Control Sheet

| | |
|---|---|
| Prepared for | 3m Architects |
| Project Name | Stanwell New Projects Office |
| Site Address | Switchyard Road |
| File Reference | Q250128-HYD-PS001 |
| Revision | C |
| Date | 19.11.2025 |

# Revisions

| Revision | Date | Description | Prepared | Reviewed | Approved | Signature |
|---|---|---|---|---|---|---|
| A | 25.08.25 | For Approval | HR | JP | Harrison Ralph | |
| B | 21.10.25 | For Approval | HR | JP | Harrison Ralph | |
| C | 28.11.25 | For Approval | HR | JP | Harrison Ralph | |

# CONTENTS

# 1. EXECUTIVE SUMMARY

This Performance Solution has been produced in respect of the proposed commercial development to be constructed on Switchyard Rd Stanwell QLD 4702. It aims to outline the proposed strategy of the Performance Solution to facilitate the installation of 4x. valley box gutters that deviate from the Deemed to Satisfy (DtS) box gutter design provisions as nominated in AS/NZS3500.3:2021.

The basis of this Performance Based Design Brief (PBDB) is to demonstrate that the performance requirements of the National Construction Code (NCC) 2022, Volume 1, has been satisfied by providing a performance-based solution.

Pursuant to Clause A2G3 (2) (b) of the NCC, this Performance Solution has been prepared by JHA Consulting Engineers QLD P/L at the request of the client and relates to the proposed subject development identified above.

### SUMMARY TECHNICAL DEPARTURE LIST

The following table is a summary of the proposed technical departures against the relevant performance requirements of the NCC.

| Item | Deemed-to-Satisfy Provision | Technical Requirement | Technical Departure |
|---|---|---|---|
| 1 | **F1D3 – Stormwater Drainage**<br>Stormwater drainage must be designed and constructed in accordance with AS/NZS 3500.3. | **AS/NZS3500.3:2021**<br>Clause 3.7.3 – Limitations relating to box gutter gradient. | The assessment of the gutter gradient<br>a) Box gutter gradient under the applicable DtS provisions of AS/NZS 3500 .3 :2021 is limited to gradients being in the range of 1:40 to 1:200<br><br>The departure relates to providing a gutter that has a gradient that exceeds the DtS maximum grade of 1:40 nominated in AS/NZS 3500 .3 :2021. |
| | | **AS/NZS3500.3:2021**<br>Clause 3.7.6 – Layout<br>(g) Box gutter shall –<br>(i) be straight (without change in direction); | The departure relates to providing a box gutter that technically changes direction, to enable the capture of a valley gutter adjacent. |
| | | **AS/NZS3500.3:2021**<br>Clause 3.6.1<br>**b)** Max catchment size of 20m2 | This departure relates to our maximum DtS catchment size exceeding the 20m2 to a valley gutter |

Table 1 - Technical Departures

## 2. PROJECT DESCRIPTION

The project is a single level commercial building to be constructed at Switchyard Rd Stanwell QLD 4702.

The building is a quadrangle based design, the proposed roof design is falling towards a central courtyard, the roof is comprised of the following;

- 4x. conventional box gutters, which are located outside of the internal building envelope below.
- 4x. unconventional valley gutters which have a greater catchment area than nominated under AS/NZS3500.3:2021
- Clause 3.6.1 and therefore, considered beyond scope of general method which necessitates the use of a box gutter profile rather than the traditional DtS "V" shaped valley gutter profile
- 4x. unconventional box gutters which incorporate a change of direction, which is not permitted under AS/NZS3500.3:2021 Clause 3.7.6 (g)(i) and is therefore, considered beyond scope of general method
    - We note that whilst there is a change in direction in the box gutter this would be not indifferent to the DtS valley gutter to box gutter if the traditional 'V' shaped valley gutter was utilized.

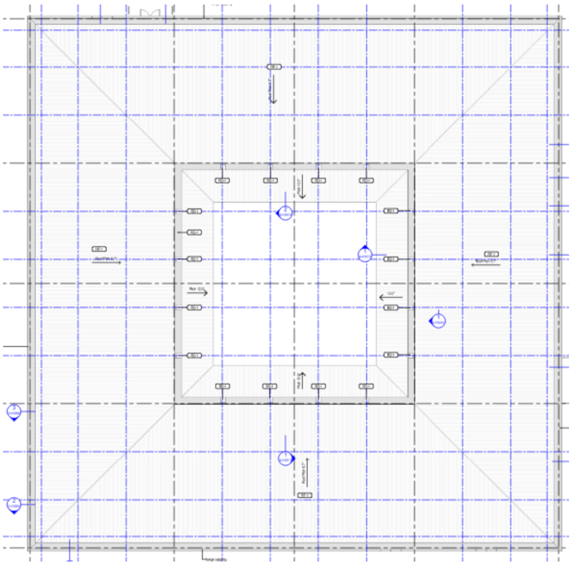


*Figure 1 - Proposed Building Image*

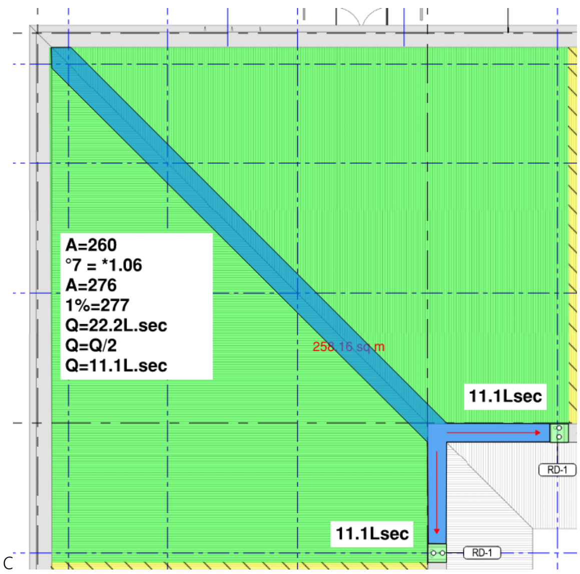



*Figure 2 – Indicative Proposed Roof Plan*

# 3. RELEVANT STAKEHOLDERS

Consultation and active engagement with relevant stakeholders are an important consideration in the process of developing the performance solution.

The following representatives, as listed in Table 2 below, have been identified as being the key stakeholders for the project.

We request that all stakeholders confirm their engagement in the production and completion of the Performance Solution Report.

Stakeholders' confirmation may be provided by the following methods:

- Signature on the table provided below
- Meeting minutes recording all the names of the stakeholders providing confirmation of their engagement in the preparation of the performance solution.
- Individual stakeholder e-mail correspondence providing confirmation of their engagement in the preparation of the performance solution.

Written confirmation is required from all relevant stakeholders confirming their engagement in the preparation of the performance solutions.

| Stakeholder | Company | Name | Signature |
|---|---|---|---|
| Property Owner | Stanwell Corporation Limited P/L | Brenton Jenson | |
| Architect | Brian Hooper Architect P/L | Brian Hooper | |
| Building Certifier | Formiga 1 P/L | Shawn Brosnan | |
| Hydraulic Engineer | JHA Consulting Engineers QLD P/L | Mr. Harrison Ralph | |

# 4. TECHNICAL DATA

## APPLICABLE LEGISLATION/CODES

This report is based in the context of:

- National Construction Code (NCC) 2022, Volume 1.
- AS/NZS 3500.3-2021 Plumbing and Drainage Part 3 – Stormwater Drainage.

## RELEVANT DTS CLAUSES

The relevant deemed to satisfy provisions that are the subject of this performance solution are:

*National Construction Code (NCC) 2022, Volume 1 – F1D3 – Stormwater Drainage*

Stormwater drainage must be designed and constructed in accordance with AS/NZS 3500.3.

## TECHNICAL REQUIREMENTS

The relevant technical requirements that are the subject of this performance solution are:

1. AS/NZS 3500 .3 :2021 Clause 3.7.3 – Box Gutters design limitations – min & max gradient

The extract below from AS/NZS 3500.3:2021 nominates the Box Gutter DtS provisions.

**3.7.3 Limitations**

The following limitations of the general method apply to box gutter systems:

(a) Gradients shall be in the range 1:40 to 1:200.

NOTE 1 Figures H.6 and H.8 assume that box gutters slope in the range 1:40 to 1:200.

(b) Rainheads —

(i) design flows shall not exceed 16 L/s; and

(ii) size range of vertical downpipes shall be according to Figure H.3.

(c) The limitation of solution for sumps with overflow devices shall be the size range of vertical downpipes according to Figure H.4.

NOTE 2 Criteria for box gutter overflow devices are given in Clause 3.7.7 and illustrated in Figure 3.7.3.

NOTE 3 The minimum width of box gutters used for commercial construction is 300 mm. In Australia, box gutters 200 mm wide may be used for domestic construction, but they are more prone to blockages. Additional height is recommended where possible.

NOTE 4 See Appendix I for examples of the application of the general method.

*Figure 3 - Extract from AS/NZS3500.3:2021 Clause 3.7.3*

As per the above, the minimum gradient of a box gutter is 1:200 and the maximum gradient is 1:40 and the proposed gutter will be 1:8 gradient.

2. AS/NZS3500.3:2021 Clause 3.7.6 (g) (i) Box Gutter layout

(g) Box gutters shall —

(i) be straight (without change in direction);

(ii) have a horizontal constant width base (sole) with vertical sides in a cross-section;

(iii) have a constant longitudinal slope between 1:200 and 1:40;

*Figure 4 - AS/NZS3500.3:2021 Clause 3.7.6 (g) (i)*

3. AS/NZS3500.3:2021 Clause 3.6.1 (c) Valley gutter Limitations

**3.6 Valley gutters**

**3.6.1 Limitations**

The following limitations of the general method apply for valley gutters:

(a) Roof slopes shall be not less than 1:4.5 (12.5°).

(b) The nominal valley gutter side angle shall be 1:3.4 (16.5°).

NOTE See Figure 3.6.1 for a profile of valley gutter.

(c) The catchment area shall not exceed 20 $m^2$.

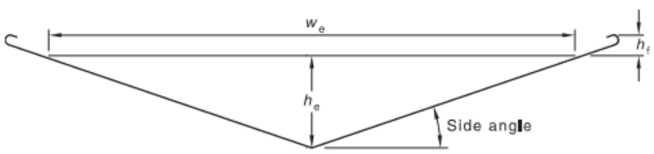



**Figure 3.6.1 — Profile of a valley gutter**

*Figure 5 - AS/NZS3500.3:2021 Clause 3.6.1 (c)*

## PERFORMANCE REQUIREMENTS

The relevant performance requirements that are the subject of this performance solution are:

### *F1P2 Preventing rainwater from entering buildings*

Surface water, resulting from a storm having an annual exceedance probability of 1%, must not enter the building.

### *F1P3 Rainwater drainage systems*

A drainage system for the disposal of surface water resulting from a storm having an annual exceedance probability of—

a. 5% must—

i. convey surface water to an appropriate outfall; and

ii. avoid surface water damaging the building; and

b. 1% must avoid the entry of surface water into a building.

#### TECHNICAL DEPARTURES

The following areas of the works have been identified as areas that do not meet the Deemed-to-Satisfy provisions of the PCA and referenced Australian Standards:

- The assessment of the maximum gradient of a box gutter under the applicable DtS provisions of AS/NZS 3500 .3 :2021 is limited to a maximum gradient of 1:40 as defined in Figure 3.7.3(a). The departure relates to providing a gutter that has a gradient that exceeds the 1:40 limitation nominated under AS/NZS 3500 .3 :2021.
- The assessment of the layout of a box gutter under the applicable DtS provisions of AS/NZS 3500.3:2021 state box gutters shall be straight (without change in direction).

  The departure relates to the providing a gutter with a change in direction.
- The assessment of the limitations of a valley gutter under the applicable DtS provisions of AS/NZS 3500 .3 :2021 is limited to a maximum catchment area of 20m2. The departure relates to the providing a valley gutter (with a box gutter profile) that exceeds a 20m2 catchment area.

# 5. PERFORMANCE SOLUTION

## RISK & CONTROL MEASURES

By increasing the gradient of the box gutter and incorporating a change in direction, we run the risk of the gutter and sump filling up faster than the water can be drained, if this happens the water could ingress into the building via the roof sheets. To prevent this from occurring, it is proposed to implement the following control measures.

1. Increase the capacity of the box gutter to ensure the hydraulic capacity is greater than that prescribed under the DtS solution increasing the redundancy.
2. Incorporating 2 rain heads each with high capacity vertical overflows, to ensure the "box-valley-gutter" can safely drain, should the primary system be blocked and to split the roof catchment area flow over all sumps.
3. Customised gutter design to direct the flow of water into each rain head ensuring the flows are balanced.

## DESIGN VALIDATION

The gutter design will be validated using a proprietary valley gutter & box gutter calculator which are based on the DtS general method provisioned under AS3500.3:2021 to provide the min. cross sectional area, additionally we will also utilise the Manning's channel calculation formula to demonstrate the hydraulic capacity of the gutter and validate the proposed valley gutter dimensions.

## MAINTENANCE

All roof drainage systems require routine maintenance to prevent foreign objects and organic matter from collecting on the roof and gutters that results in reduced gutter performance and eventually blockages and leaks. Routine maintenance should be scheduled as a minimum between every seasonal transitional period and post major weather events.

## DESIGN METHODOLOGY

This performance solution will be broken into two parts; part one will address the valley gutter design and part two will address the box gutter design.

### BOX GUTTER DESIGN

The relevant performance requirements for the box gutter will be addressed via the following solution. We will utilise DtS design methodology to demonstrate the gutter is sized with redundancy.

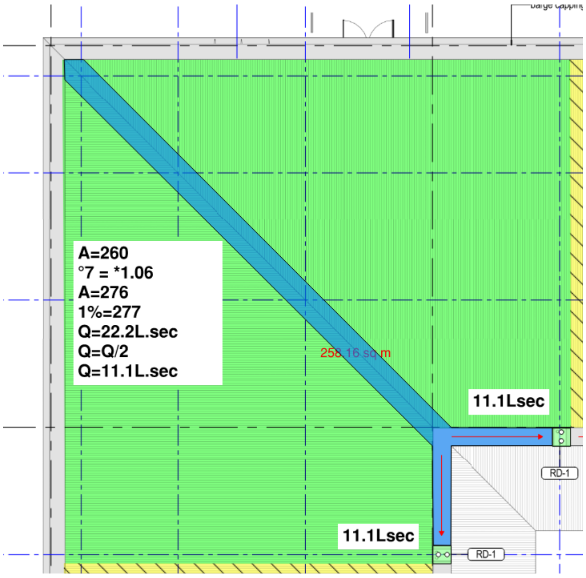



*Figure 6 Proposed Valley Box Gutter*

- Determine 1% AEP for 5 min duration = 277mm/hr
- Determine roof catchment areas
  - LHS 130sqm + RHS 130sqm
  - Apply roof slop multiplier factor for 7° slope = *1.06
  - LHS 138sqm + RHS 138sqm
  - Calculate gutter flow rate (A*1%AEP/3600) = 22.2Lsec (Valley gutter design flow rate)
  - Calculate gutter flow rate to each rain head (Q/2) = 11.1L/sec (Box gutter design flow)
  - Calculate min. gutter dims, for this we use a paid subscription for roof gutter designs by Ken Sutherland, refer below and appendix for further information.

    Upper Gutter (Valley Gutter)

    - Box gutter Width 600mm
    - Box gutter depth 95mm increased to 200mm
    - Gradient 1:8 or 12%

    Lower Gutter (main roof level box gutter)

    - Box gutter Width 600mm
    - Box gutter depth 95mm increased to 250mm
    - Gutter slop 1:200 or 0.5%

    Sump Dimensions

    - Sump width 600mm

- Sump depth 150mm
- Sump length 800mm

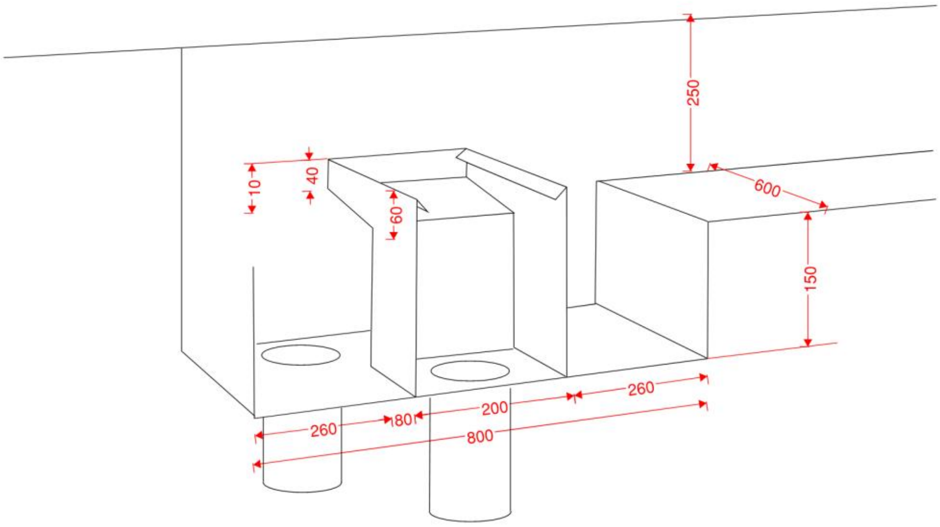



*Figure 7 – DtS vs PS Box Gutter Detail*

## MANNINGS FLOW CALCULATION

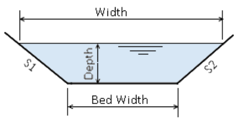
In order to provide further assurance, the box gutter is suitably sized and will not overflow in a significant storm event, we will calculate the depth of water using the Mannings flow calculation formula.

Q = A/n * R^0.66 * S^0.5

Where:-
Q = Flow
A = Cross sectional area of the water flow
R = Hydraulic radius = A/P
P = Wetted Perimeter
S = bed slope

**Enter Details**

| | |
|---|---|
| Flow (L/s) | 11.1 |
| Slope S1, vertical to horiz (1:?) | 1 |
| Slope S2, vertical to horiz (1:?) | 1 |
| Bed Slope (1:?) | 8 |
| Bed Width (mm) | 600 |
| Mannings 'n' | 0.013 |




Mannings 'n' (Typical Values)
- Mowed Grass - 0.025
- Concrete - 0.015
- Asphalt - 0.016
- Rocks - 0.035

Press to Calculate (or press 'Enter')

**Channel Size**

Water Surface Width (mm) 626 Depth (mm) 13.000( Flow Velocity (m/s) 1.47

*Figure 8 Mannings Open Channel Calculation – Upper Gutter Test (valley gutter)*

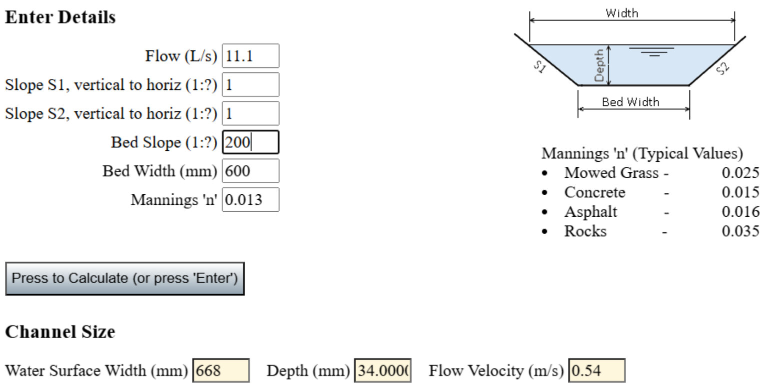



*Figure 9 Mannings Open Channel Calculation – Lower Gutter Test (main building level box gutter)*

Based on this exercise we confirm the following redundancy is achieved.

Upper gutter (valley gutter)

- Gutter 600mm(W) x 200mm (D)
- Depth of water 13mm
- Free board 50mm
- Redundancy 58% (mannings vs refrenced design)

Lower gutter (main building level box gutter)

- Gutter 600mm(W) x 200mm (D)
- Depth of water 34mm
- Free board 50mm
- Redundancy 44% (mannings vs refrenced design)

*Figure 10 Box Gutter Free Board Detail*

# 6. SUMMARY

To summarize this performance solution, highlights a deviation from the AS3500.3:2021 DtS provisions by increasing the slope of the box gutter and addressed this by increasing the capacity of the gutters.

Based on the above assessment, it is our professional opinion that the proposed rainwater system which consists of;

1. Upper Gutter - 600mm wide by 200mm deep valley gutter
2. Lower Gutter – 600mm wide by 250mm deep box gutter
3. Sump - 800mm long by 150mm deep by 600mm wide rain head and 2x. 150Ø downpipes

Th above components work together to achieve compliance with the following performance requirements nominated in National Construction Code (NCC) 2022, Volume 1:

- F1P2 Preventing rainwater from entering buildings
- F1P3 Rainwater drainage systems

Harrison Ralph

Associate

# 7. APPENDIX A – HYDRAULIC DESIGN CALCULATIONS

Stanwell Box Gutter Number 1 Q250128

**BOX GUTTER AND VERTICAL OVERFLOW DESIGN TO AS/NZS 3500.3:2021**

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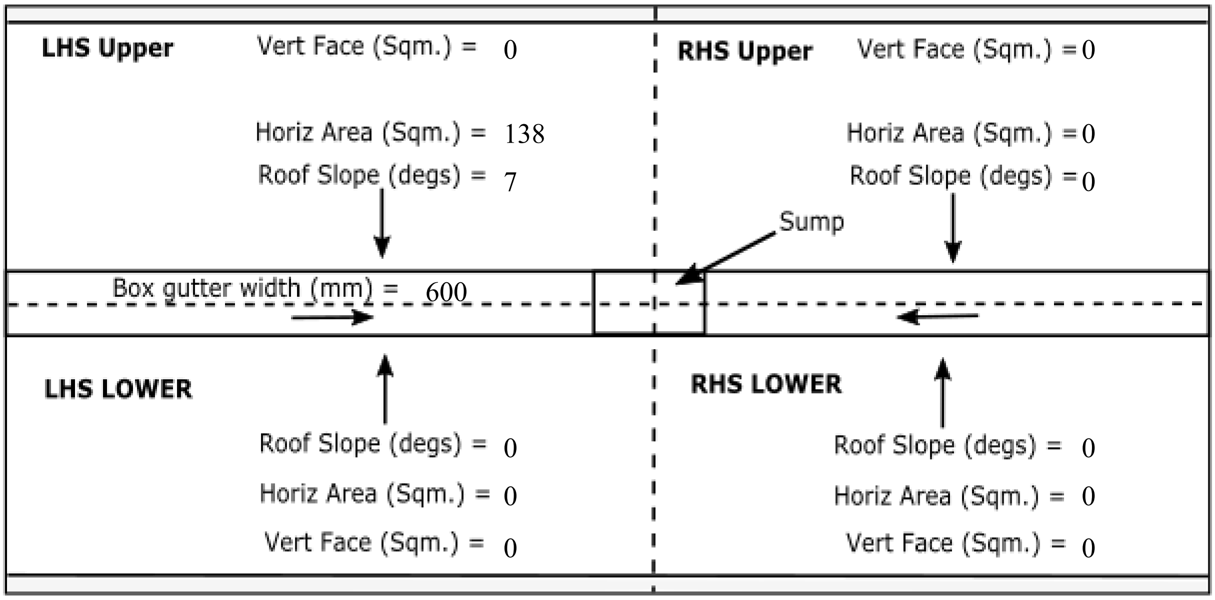
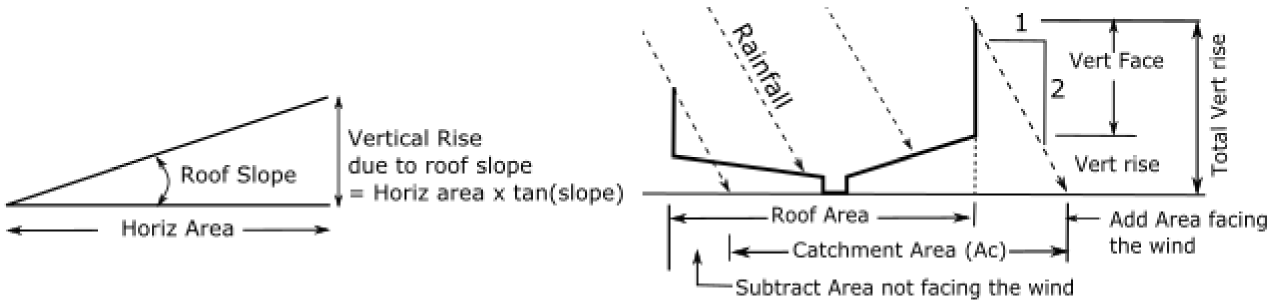


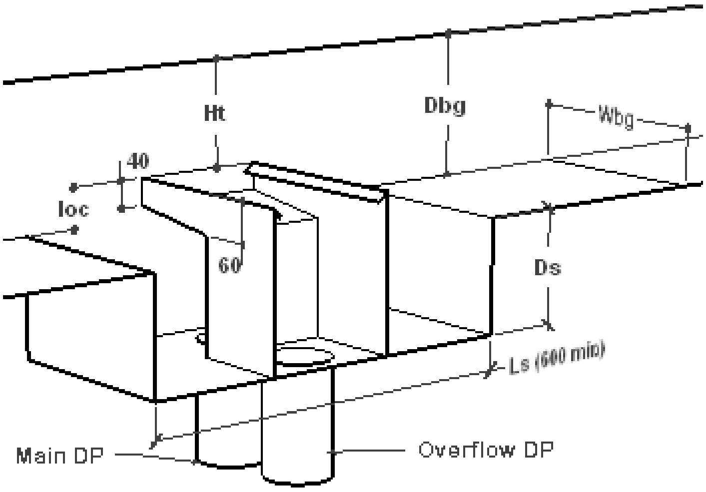



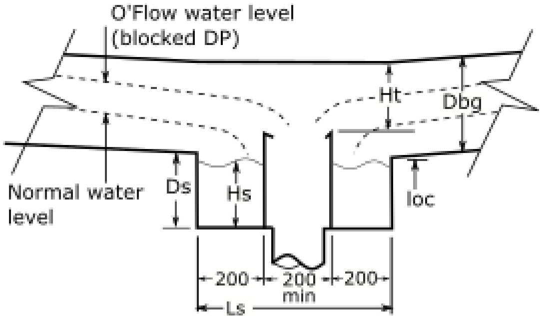

**Calc Total Vertical rise (Av)**

Total Vertical rise (Av) = [Roof horiz Area * tan(roof slope)] + (Area of Vert face)

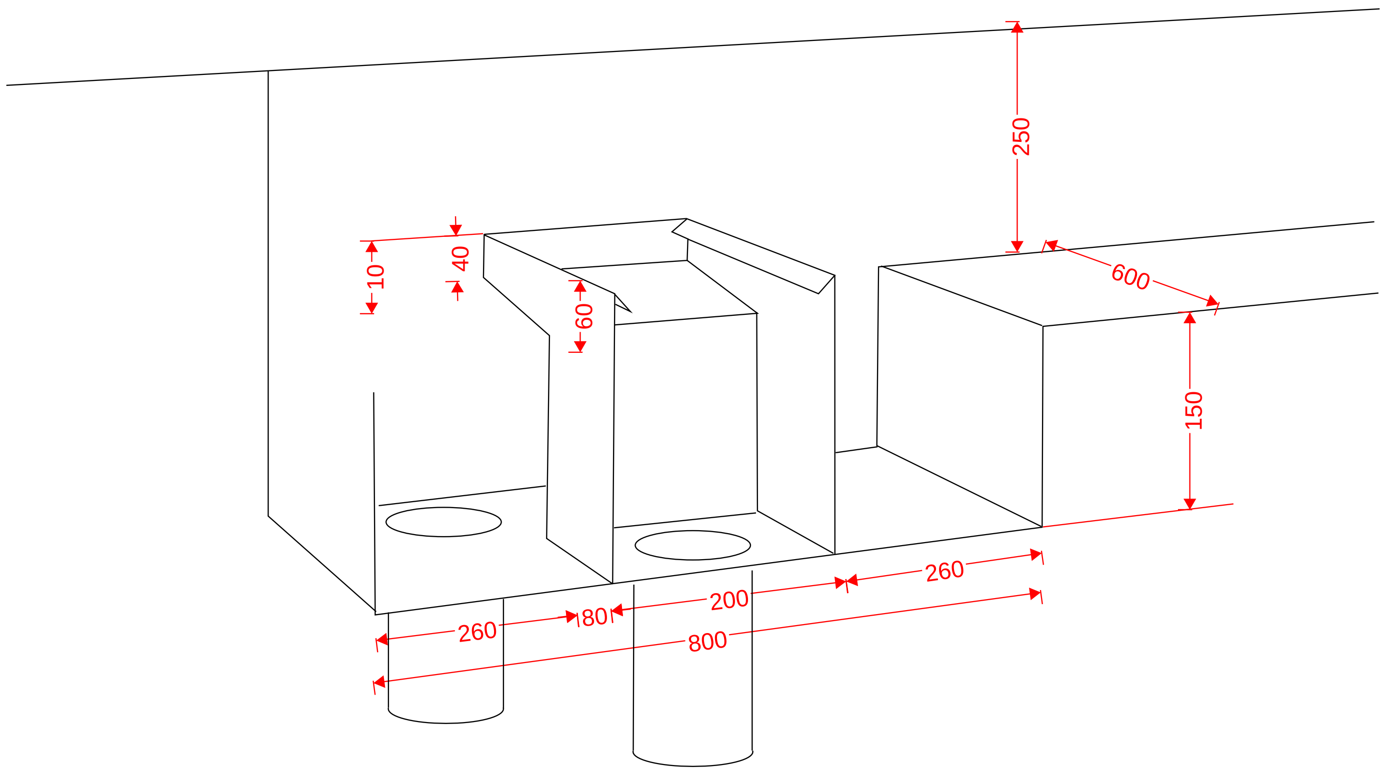
| | | | | |
|---|---|---|---|---|
| Total Vert Rise area LHS upper = [138 * tan(7)] + 0 | Av_Lhs_u | = | 16.9 | sq.m |
| Total Vert Rise area RHS upper = [0 * tan(0)] + 0 | Av_rhs_u | = | 0 | sq.m |
| Total Vert Rise area for all upper = 0 + 16.9 | Av_u | = | 16.9 | sq.m |
| Total Vert Rise area LHS lower = [0 * tan(0)] + 0 | Av_Lhs_L | = | 0 | sq.m |
| Total Vert Rise area RHS lower = [0 * tan(0)] + 0 | Av_rhs_L | = | 0 | sq.m |
| Total Vert Rise area for all lower = 0 + 0 | Av_L | = | 0 | sq.m |

| | | | |
|---|---|---|---|
| **Calculate Total Horizontal areas (Ah)** | | | |
| Total Horiz area for all LHS = 138 + 0 | Ah_Lhs | = 138 | sq.m |
| Total Horiz area for all RHS = 0 + 0 | Ah_rhs | = 0 | sq.m |
| **Find worst wind direction** | | | |
| = Largest vertical Area facing the wind | | | |
| = The larger of (Av_u) and (Av_L) = Av_u | | = 16.9 | sq.m |
| worst wind direction (blows on this face), therefore wind direction | | = from Lower | sq.m |
| **Find Cment area LHS with wind from Lower** | | | |
| = Ah_Lhs + 1/2(Av_Lhs_u - Av_Lhs_L) | Ac_LHS | = 146.45 | sq.m |
| **Find Cment area RHS with wind from Lower** | | | |
| = Ah_rhs + 1/2(Av_rhs_u - Av_rhs_L) | Ac_RHS | = 0 | sq.m |
| Design cment area for box gutter worst case being Ac_LHS | Ac_BG | = 146.45 | sq.m |
| Design cment area for Sump and DP | | | |
| =Ac_LHS + Ac_RHS | | = 146.45 + 0 | |
| | Ac_DP | = 146.45 | |
| **Calculate Design Flows** | | | |
| Design Storm Intensity (ARI 100) | I | = 277 | mm/hr |
| Design Flow for BG = (Int * Area) / 3600 | | = (277 * 146.45)/3600 | L/sec |
| | Qbg | = 11.3 | L/sec |
| Design Flow for DP and Sump = (Int * Area) / 3600 | | = (277 * 146.45)/3600 | L/sec |
| | Qdp | = 11.3 | L/sec |




| | | | | |
|---|---|---|---|---|
| Box gutter width | Wbg | = | 600 | mm |
| Box gutter slope | | = | 1:200 | |
| From Fig H1 Depth for free flow conditions | Ha | = | 114 | mm |
| down pipe and Oflow pipe size | dia | = | 150 | mm |
| from Fig H4 Theoretical Sump Depth | Hs | = | 107 | mm |
| from Fig H6(a) loc | loc | = | 34 | mm |
| from Fig H8 Ht | Ht | = | 91 | mm |
| loc + Ht | | = | 125 | mm |
| BG depth is the max of Ha and (loc+Ht) | Dbg | = | 125 | mm |
| loc < 60 therefore From Note1(a) Fig H7 Datum for sump depth is D/S sole of OFlow channel | Ds | = | Hs + (60-loc) | mm |
| | Ds | = | 133 | mm |
| Sump depth < min allowable of 150 therefore | Ds | = | 150 | mm |
| Min length of sump adopted | Ls | = | 600 | mm |


250
10
40
60
600
150
260
180
200
260
800

## VALLEY GUTTER DESIGN USING COMPUTATIONAL FLUID DYNAMICS (CFD)

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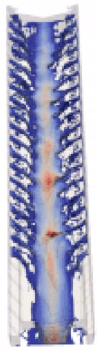
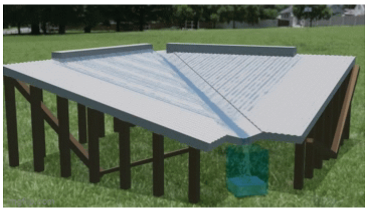


### The Method

This programme has been derived from hundreds of calculations with a computational fluid dynamics (CFD) programme.
The programme used was FLOW-3D Hydro.



A range of flows from 1 L/s to 100 L/s was used.
And a range of slopes from 5 degrees to 35 degrees was also used.

This huge number of CFD results was sufficient to plot charts and calculate equations for over a 150 different cases.

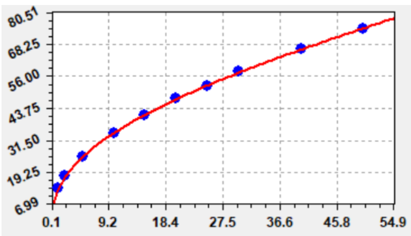
The programme uses the resultant equations to calculate valley gutter sizes for anything within the above range of flow and roof slopes.

If the desired roof slope does not fall on a calculated case the programme will interpolate.
As you can see with the Roof Slopes chart , it is very easy to interpolate between the different slopes.

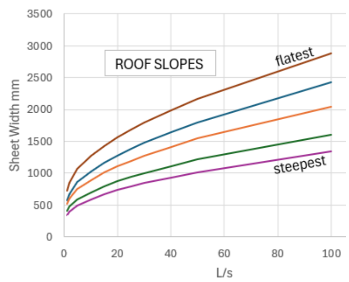
Find CFD Depth from flow

| Roof Slope | 23.7 deg | Side Slope | 16.51227 | |
|---|---|---|---|---|
| Flow L/s | CFD Depth mm | Dth from eqn (mm) | difference | % |
| 1.111 | 13.12 | 13.10 | 0.02 | 0.117094 |
| 2.222 | 17.83 | 17.71 | 0.12 | 0.677624 |
| 5 | 25.4 | 25.26 | 0.14 | 0.540534 |
| 10 | 34.09 | 34.40 | -0.31 | -0.916239 |
| 15 | 41.17 | 41.38 | -0.21 | -0.515938 |
| 20 | 47.57 | 47.31 | 0.26 | 0.539512 |
| 25 | 52.18 | 52.61 | -0.43 | -0.823106 |
| 30 | 58.03 | 57.48 | 0.55 | 0.951852 |
| 40 | 66.42 | 66.37 | 0.05 | 0.079302 |
| 50 | 74.38 | 74.53 | -0.15 | -0.199608 |

**CFD depth vs Flow table**
for 23.7deg roof slope.




**Graph of Flow table**
for 23.7 deg roof




**Comparison of roof slopes**

NOTE: For roof slopes greater than 25deg, the roof sheeting protrudes into the running water.
To prevent this obstructing the flow, an extra free board of 15mm has been applied to all roof slopes steeper than 25 degrees.
This has the effect of increasing the sheet width.

## CALIBRATION

Once all the equations were determined the resultant programme was calibrated against table 3.6.2 in AS/NZS 3500.3 as shown below.

**Table 3.6.2 — Valley gutters — Dimensions**

| Design rainfall intensity mm/h | | Minimum, mm | | |
|---|---|---|---|---|
| | | Sheet width | Effective depth ($h_e$) | Effective width ($w_e$) |
| | ≤ 200 1.111 L/s | 355 | 32 | 215 |
| > 200 | ≤ 250 1.4 L/s | 375 | 35 | 234 |
| > 250 | ≤ 300 1.66 L/s | 395 | 38 | 254 |
| > 300 | ≤ 350 1.94 L/s | 415 | 40 | 273 |
| > 350 | ≤ 400 2.222 L/s | 435 | 43 | 292 |

NOTE 1 This Table is derived from Martin and Tilley.

NOTE 2 Freeboard ($h_f$), 15 mm.

NOTE 3 The sheet width from which the valley is to be formed has been calculated on the basis of $h_f$ = 15 mm and an allowance for side rolls or bends of 25 mm.

Table 3.6 .2 has a number of givens:-
The catchment area is 20 square metres. The roof slope is 23.5 degrees, and the side slope is 16.5 degrees, and the freeboard is 15mm.

NOTE: To get a side slope of 25 degrees, it requires a roof slope of 23.7.
However the difference in slope is so minor it does not affect the result.

From these figures the runoff is calculated as follows:
Runoff (Q)= Rainfall intensity(I) * Catchment Area (A) / 3600 ;
Q = I * A/3600

In table 3.6.2 above, for intensity 200mm/h and catchment area 20 sqm,
runoff (Q) = 200*20/3600 L/s = 1.111L/s
Similarly for intensity of 400mm/hr Q = 2.222L/s

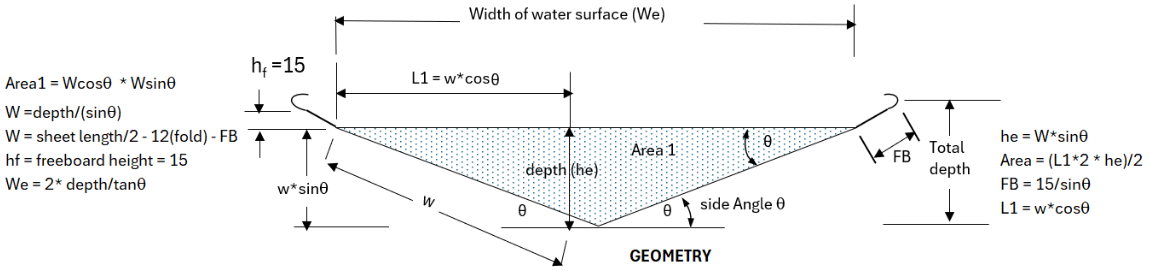
As can be seen from the table of CFD flow depths for a 23.7 deg roof, the flow depths for a flow of 1.111 L/s has a calculated depth = 13.10 and 2.222 L/s has a calculated depth = 17.71 mm

These are a lot less than the effective depth for these flows as shown in table 3.6.2.

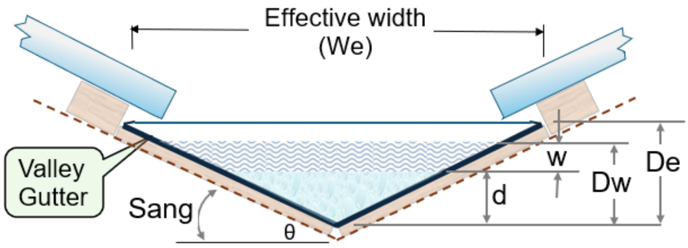
This difference is made up of:-
a) the allowance for wave motion, and
b) a free board of 15mm.

Find CFD Depth from flow

| Roof Slope | 23.7 deg | Side Slope | 16.51227 | |
|---|---|---|---|---|
| Flow L/s | CFD Depth mm | Dth from eqn (mm) | difference | % |
| 1.111 | 13.12 | 13.10 | 0.02 | 0.117094 |
| 2.222 | 17.83 | 17.71 | 0.12 | 0.677624 |
| 5 | 25.4 | 25.26 | 0.14 | 0.540534 |
| 10 | 34.09 | 34.40 | -0.31 | -0.916239 |
| 15 | 41.17 | 41.38 | -0.21 | -0.515938 |
| 20 | 47.57 | 47.31 | 0.26 | 0.539512 |
| 25 | 52.18 | 52.61 | -0.43 | -0.823106 |
| 30 | 58.03 | 57.48 | 0.55 | 0.951852 |
| 40 | 66.42 | 66.37 | 0.05 | 0.079302 |
| 50 | 74.38 | 74.53 | -0.15 | -0.199608 |



**Analysing table 3.6 .2**
Taking the last row of the table
where flow = 2.222 L/s,
Calibrate for effective depth:




| | | | | |
|---|---|---|---|---|
| **Effective depth from table 3.6.2** | **De** | = | **43** | mm |
| Freeboard | FB | = | 15 | mm |
| Depth with wave (De-FB) | Dw | = | 43-15 | mm |
| | | = | 28 | mm |
| CFD depth | d | = | 17.71 | mm |
| Therefore wave motion | W | = | 28-17.71 | mm |
| | | = | 10.29 | mm |
| Depth adjusted for wave motion | Dw | = | 17.71 + 10.29 | mm |
| | | = | 28 | mm |
| Effective depth (Add FB) | De | = | 28 + 15 | mm |
| Effective depth | De | = | 43 | mm |

**This equals the effective depth from table 3.6.2**

| | | | | |
|---|---|---|---|---|
| Calculate (W) ratio to (d) | | = | 10.29 / 17.71 | mm |
| | | = | 0.58 | |

So, for all other cases, depth including wave motion (Dw)
equals CFDdepth + 0.58*CFDdepth

**Calibrate for Effective width (We)**
Taking the last row of the table where flow equals 2.222 L/s

| | | | | |
|---|---|---|---|---|
| **Effective width from table** | **We** | **=** | **292** | mm |
| From geometry | We | = | 2 * De/tan(side slope) | |
| | We | = | 2 * 43/tan(16.5) | |
| | We | = | 2 * 145 | |
| **Therefore Effective width** | **We** | **=** | **290 (near enough)** | |

**Calibrate for Sheet width (SW)**
Taking the last row of the table where flow = 2.222 L/s)

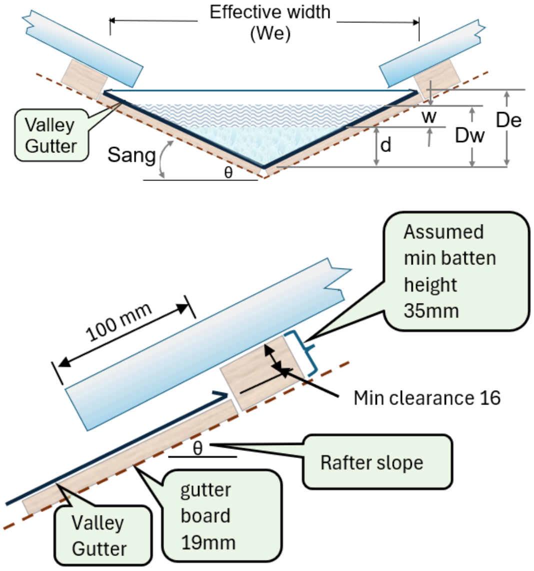
| | | | | |
|---|---|---|---|---|
| **Sheet width from table** | **SW** | **=** | **435** | mm |
| From geometry, side width | w | = | depth(De)/sine(side angle) | |
| | w | = | 43 / sin(16.5deg) | |
| | w | = | 43/0.284 | mm |
| | w | = | 151.4 | mm |
| From geometry, FB length on slope | FB | = | 15/sin(side angle) | |
| | w | = | 15 / 0.284 | mm |
| | w | = | 52.8 | mm |

Therefore SW = 2*w + 2*FB + Allowance of 25 for end curls
= 2*151.4 + 2*52.8 + 25
= **433.4 mm (near enough allowing for rounding)**

**THIS PROJECT**
Using the above methodology, the following are the results for this project.

Note that the roof sheeting will start obstructing the flow for all slopes greater than 25 deg.
If this is the case, an extra freeboard of 15mm has been added.
This also increases the sheet width.

| Parameter | Sym | Unit | Value |
|---|---|---|---|
| Catchment Area | CA | sq.m | = |
| Rainfall Intensity(1%AEP 5min) | Int | mm/hr | = |
| Roof Slope | RS | degs | = |
| Flow | Q | L/s | = |
| Gutter Slope | GS | degs | = |
| Gutter Side Angle | Sang | degs | = |
| CFD Depth | d | mm | = |
| Allowance for Wave Motion | w | mm | = |
| Depth Adjusted for Wave Motion | Dw | mm | = |
| Total Effective Depth (incl. FB) | De | mm | = |
| Effective Width | We | mm | = |
| Required Sheet Width | SW | mm | = |



**Valley gutter arrangement used in this programme**

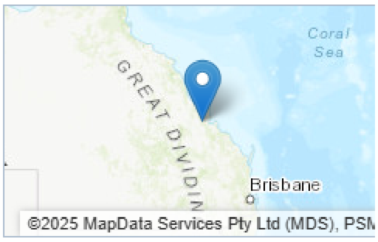
# 8. APPENDIX B – BOM IFD TABLE

## Location

**Label:** Not provided

**Latitude:** -23.5072 [Nearest grid cell: 23.5125 (S)]

**Longitude:** 150.319 [Nearest grid cell: 150.3125 (E)]




## IFD Design Rainfall Intensity (mm/h)

Issued: 22 August 20

Rainfall intensity for Durations, Exceedance per Year (EY), and Annual Exceedance Probabilities (AEP
FAQ for New ARR probability terminology

Table | Chart

Unit: mm/h

| | Annual Exceedance Probability (AEP) | | | | | | |
|---|---|---|---|---|---|---|---|
| **Duration** | **63.2%** | **50%#** | **20%*** | **10%** | **5%** | **2%** | **1%** |
| **1 min** | 154 | 171 | 223 | 260 | 296 | 344 | 382 |
| **2 min** | 129 | 144 | 189 | 220 | 250 | 290 | 321 |
| **3 min** | 122 | 135 | 178 | 207 | 235 | 273 | 302 |
| **4 min** | 117 | 129 | 170 | 197 | 225 | 261 | 289 |
| **5 min** | 112 | 124 | 163 | 189 | 215 | 250 | 277 |